

 MERIDIAN FOUNDATION CLINICAL INVESTIGATOR GRANT PROGRAM (FY2026–2029)

Targeted Nanoparticle Immunotherapy in Refractory Triple-Negative Breast Cancer: Phase IIa Clinical Trial & Biomarker Cohort Analysis

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Proposed Project Period: Oct 1, 2026 — Sep 30, 2029 (36 Months)
Total Requested Budget: \$1,850,000 USD
IRB Approval Status: Approved (#2026-VAMC-0492)
FDA IND Status: Active (IND #168,402)
Primary Study Site: Vertex Clinical Research Center

1. Project Summary & Specific Aims

1.1 Executive Summary

Project Synopsis: Triple-Negative Breast Cancer (TNBC) accounts for 15–20% of all invasive breast carcinomas and exhibits disproportionate mortality due to rapid disease progression, early metastatic recurrence, and a lack of targeted therapeutic options. While immune checkpoint blockade (ICB) has improved outcomes in PD-L1-positive disease, over 65% of refractory TNBC patients fail to achieve durable clinical responses. This proposal presents a multi-center Phase IIa clinical trial investigating **Nexa-NP402**—a novel lipid-polymer nanoparticle encapsulating STING agonists alongside anti-PD-1 monoclonal antibodies. By converting immunologically “cold” tumor microenvironments into highly inflamed “hot” sites, this intervention aims to overcome primary therapeutic resistance. Supported by robust preliminary pre-clinical data from Synapse Bio, this 3-year study evaluates clinical efficacy, rigorous patient cohort power modeling, and longitudinal single-cell spatial transcriptomics in 96 stratified patients.

1.2 Background & Clinical Rationale

Metastatic TNBC carries a dismal 5-year overall survival rate of under 12%. Current standard-of-care regimens relying on systemic platinum-based chemotherapy yield transient responses burdened by severe cumulative toxicity. Recent clinical trials demonstrate that monotherapy with anti-PD-1 agents achieves objective response rates (ORR) of only 10–15% in unselected refractory populations. The primary mechanistic bottleneck is the immunosuppressive tumor microenvironment (TME), characterized by dense tumor stroma, defective antigen presentation, and a scarcity of tumor-infiltrating CD8+ T lymphocytes.

Nexa-NP402 is engineered to target tumor-associated macrophages (TAMs) via mannose-receptor endocytosis, delivering localized cyclic dinucleotides (STING agonists) while simultaneously blocking the PD-1/PD-L1 axis. In preliminary murine models conducted at Vertex AMC, Nexa-NP402 elicited complete tumor regression in 58% of subjects and established long-term immunological memory without inducing systemic cytokine release syndrome.

1.3 Hypothesis & Specific Aims

We hypothesize that co-administration of Nexa-NP402 with pembrolizumab will significantly prolong progression-free survival (PFS) in patients with refractory TNBC by reprogramming the TME and driving clonal expansion of tumor-reactive T cells.

[Aim 1]Phase IIa Clinical Efficacy & Safety Evaluation: Conduct an open-label, randomized 1:1 Phase IIa trial ($N = 96$) comparing Nexa-NP402 + pembrolizumab against physician’s choice single-agent chemotherapy. Primary endpoint: Progression-Free Survival (PFS) at 12 months. Secondary endpoints: Overall Response Rate (ORR) per RECIST v1.1 and safety profile per CTCAE v5.0.

[Aim 2]Patient Cohort Power Analysis & Biomarker Profiling: Perform high-dimensional single-cell RNA sequencing (scRNA-seq) and spatial transcriptomics on pre- and post-treatment core biopsies. Evaluate baseline immune signatures, interferon-gamma gene response scores, and somatic mutation burden as predictive biomarkers of response.

[Aim 3]Pharmacokinetics & Circulating Tumor DNA (ctDNA) Dynamics: Track serial liquid biopsies at baseline, Week 2, Week 6, and progression to quantify ctDNA clearance kinetics using custom targeted ultra-deep sequencing, correlating clearance rates with durable clinical benefit.

2. Study Design & Patient Cohort Power Analysis